

Optimizing Assessment Placement and Curriculum Structure through Graph-Theoretic Analysis

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Abstract— The assurance of student learning is a crucial concern in contemporary education. Critical courses have a significant impact on how well university students do and, eventually, on graduation rates. Therefore, a key goal of decision-makers at universities should be to identify such courses. This article uses graph theory visualization and metrics to study existing curriculum structure. Gephi Software is used to visualize the courses as a network graph that shows, the courses, their prerequisites, generate course complexity and identify crucial courses of the curriculum. From the produced network, graph theory metrics is applied to calculate the courses complexity and cruciality and to identify the appropriate locations for assessment components or curriculum restructure. These techniques aid administrators and faculty to restructure the curricula and create an assessment scheme. In this work academic curriculum is treated as complex system where courses are represented by nodes and directed links between each course and their prerequisites based on course catalogue. This graph provides desirable features to analyze and visualize the course using metrics of mathematical graph theory. This work aims to explore the possibility to restructure the curriculum and provide assessment for critical courses to enhance graduation rates.

Keywords—complex network, cruciality, curriculum network, education, graph theory, education

I. INTRODUCTION

Curriculum management is necessary over the duration of a university program. Understanding how vital particular courses and concepts are in a program can be informative. Considering that this can indicate which topics and courses require further attention and learning monitoring actions [1].

A subfield of mathematics known as "graph theory" examines the relationships, composition, and characteristics of graphs. A graph is essentially an object made up of vertices (nodes) and connecting lines (edges). Precisely two vertices are connected by each edge. A directed or undirected graph is possible. An undirected graph is one in which the connected vertices are linked by a line (edge) without any particular ordering. A directed graph is one in which the edge connections indicate a certain direction or ordering. This implies that one of the pair's vertices happens first and the other vertex happens second. A weighted graph has edges

and/or vertices that are assigned a value. The degree of a vertex inside a network is determined by how many edges are directly connected to it; a vertex that has a degree of 0 is referred to as being isolated [2, 3].

The first known use of graph theory dates back to a paper written by Leonhard Euler in 1736 about a method for crossing all seven bridges in Königsberg, Prussia [4]. The requirement for this challenge was that each of the seven bridges had to be crossed exactly once. Graph theory was established as a result of Euler's computational solution and abstract depiction of the issue as vertex pairs linked by edges. Since the publication of his groundbreaking work, the use of graph theory has spread to a wide range of disciplines, including education, social network analysis, disease transmission, circuit design, physics, chemistry, and biology [5].

Rollande et al. [6] utilized an extensive approach to analyze the curriculum structure by four graphs to generate students' study plan. Rollande used graph theory to generate personalized study plans for students. Aldrich [7] studied the links among the courses based on the curriculum structure to understand the university program structure. The focus of Aldrich work is to analyze the prerequisites of the courses to understand how these courses are related and use the information to take actions. Pavlich-Mariscal et al. [8] studied the courses and their learning goals where the courses, topics, and the learning goals are modeled as nodes and the prerequisite dependencies are modeled as edges. Graph analysis approaches were used to analyze the curriculum structure to automate the curriculum design. Additionally, Lightfoot et al. [9] suggested using graph-oriented approaches to address important problems like the placement of examinations, data collecting, and corrective measures within the curriculum in order to assist teachers and administrators who are tasked with developing quality assurance and assessment schemes. The focus of Lightfoot work is therefore on how to analyze program contents quantitatively and systematically. Moreover, de Blasé et al. [10] proposes a decision model using network analysis to guide curricular design committees in course selection and sequencing in higher education contexts, with a focus on the mathematics degree at the Universidad Rey Juan Carlos (URJC) in Spain. The methodology analyzes the local and global properties of the study plan, including the detection of communities within

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the graph, centrality of courses, and their dependencies, to ensure the coherence and success of the study plan. O'Meara, [11] proposes that education should be treated as a complex system of interconnected ideas, and teachers should help students make connections between these ideas to personalize learning and foster creativity. The authors use a mathematical approach to analyze the curriculum of a precalculus course, mapping textbooks into networks of concepts and using network metrics to gain insights into effective teaching and learning strategies. The work done Varagnolo [1] proposes a method to analyze the coherence of existing curricula in higher education institutions, with a focus on engineering programs at universities.

This work introduces a novel approach to the analysis and improvement of academic curriculum structures. In this contribution, we employ graph theory visualization and metrics to assess the existing curriculum and identify crucial courses. Additionally, we calculate course complexity and cruciality using graph theory metrics, aiding in the identification of suitable locations for assessment components or curriculum restructuring. Furthermore, we treat the academic curriculum as a complex system, representing courses as nodes and establishing directed links between each course and their prerequisites based on the course catalog. Within this framework, we explore opportunities to restructure the curriculum and provide targeted assessment for critical courses with the ultimate goal of enhancing graduation rates. This work significantly enhances the curriculum structure by identifying courses in need of attention and utilizing graph theory as a powerful tool for improving graduation rates.

In this article a strategy is proposed based on graph theory concepts such as analyzing node connectivity, node centrality, and the network flow which shows how courses are related within a program. Graph theory was used to enhance the curriculum structure, identify the critical courses that require further attention, and improve the graduation rate.

II. METHODOLOGY

Graph theory is a broad area of study with a diverse range of studies and applications. This paper will only take into account the comparatively limited subset of graph theory that is linked to clustering density, structural centrality, and degree. These metrics are especially relevant to the issue of curriculum design since they reflect the essential structural components required to choose where to introduce a concept, reinforce it, and assess it. The rest of this section illustrates the basic graph theory metrics applied in this work.

A. Degree of a Vertex

A vertex's degree is just a count of the edges that are linked to that vertex. Vertices with a high degree from an analysis perspective are an indication of important locations of exposure for whatever is passing in the network and provides the potential to impact or enhance the network flow. It is also an indicator of the vertex effect or significance inside a structure [2].

In an undirected graph, there is only one degree measure. There are two ways to express degree when a graph's edges are directed: in-degree and out-degree. According to their names, in-degree and out-degree both count the number of edges that enter and exit a vertex, respectively. While out-

degree analysis suggests the flow's origin, study of the in-degree measure reveals its destination [12].

B. Measures of Centrality

The concept of graph centrality is crucial to the research of graph structure since it indicates areas of influence, importance, and prominence [13, 14]. This is especially true in organizational graph structures when some type of flow is suggested [15]. What is the most significant node in the graph? is the main factor taken into account by centrality measures. What determines "important" depends on the definition used in the question [2]. Betweenness centrality is used in this study. It is used to address a distinct structural issue.

Betweenness centrality is utilized to identify the nodes or vertices that works as a bridge among two different parts of the graph. Eigenvector centrality is used as indirect measure of centrality. A major distinction among both types of centrality is that eigenvector centrality considers that some links are more critical than others [2, 16]. As a result, it takes into account both the degree of the vertex and the degree of all the vertices that are connected to it [12, 17].

C. Longest Path (Delay Factor)

The longest path for a node i is measured by the number of nodes of the longest path passing via node i .

D. Connectivity (Blocking Factor)

The total number of nodes connected to node i is what is used to determine a node's connectivity.

III. GRAPH THEORY FOR CURRICULUM ASSESSMENT AND DESIGN

The educational program curriculum can be presented as a graph which consists of nodes (vertices) and edges where various graph theory matrices can be applied to analyze the graph and enhance the curriculum structure. For the curriculum structure, courses are represented as nodes and the relationship between these courses is presented by directed edges indicating the prerequisites for each course. Nodes (courses) that are not prerequisite for any other courses will be isolated. A directed graph will be used as a prerequisite relationship that has a specific ordering. Assuming that the course is passed, typical curricula do not mandate that a course should be taken many times in order to graduate. This implies that there would be no self-loops and only one edge incident to any two vertices. As a result, the curricular graph that was produced would be categorized as simple and acyclic. These definitions imply that an academic curriculum can be visualized as a straightforward acyclic directed network.

The numbered nodes in Fig. 1. stand in for the courses' vertices. The numbering and coloring system was chosen because it shows the progress from freshman level courses to senior level courses. Therefore, the course represented by vertex 5.0 has prerequisites of 2.3, and 3.4. Based on a graph metric standpoint, vertex 2.1 has an out-degree of 2 and an in-degree of 1.

The graph in Fig. 1. can be transformed into a mathematical form that enables computation and calculations. The handling of these calculations and the provision of a visual representation of the generated graph are the focus of numerous software packages. In this work, Gephi software was utilized which is an open-source software for analyzing and visualizing large networks graphs. The free Gephi software was chosen because of its user-friendly interface and

built-in features. Additionally, the software offers a visualization feature that enables users to view the produced graph as a picture and automatically does the computations required for basic graph analysis. The software can automatically change certain aspects of this image (such as the vertex size, the edge color, and edge patterns) to show the results of standard graph metric computations. Based on the graph-theoretic metrics of a curriculum's graph, this significant analytical tool enables the user to graphically emphasize the major structural components and crucial connections. The rest of this section illustrates how this simple curriculum graph is analyzed by the graph metrics described in methodology section.

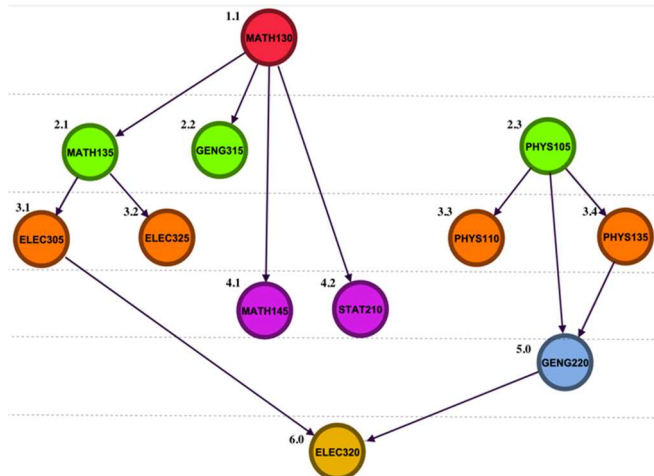


Fig. 1. Fig. 1. Sample of the electrical engineering curriculum at UAEU generated via Gephi software. Color code: term1-red, term2-green, term3-orange, term4-purple, term5-blue, and term6-yellow.

A. Curriculum Assessment

The curriculum structure displayed in Fig. 1. is analyzed via graph theory metrics to identify the crucial courses. Finding out these courses help faculty and administrators to determine the appropriate time at curriculum to introduce framework concepts, and the best places to perform baseline assessment and introduce topics.

1) *Out-Degree*: The total number of directed edges that leave a vertex is measured by the vertex's out-degree. The metric identifies the source of flow within the directed graph and provides potential areas for enhancing or influencing that flow. When used in the context of a curriculum, a vertex that has a high out-degree denotes a course that is ideally situated to introduce students to foundational knowledge or framework topics. Additionally, it recommends a suitable course for conducting baseline assessments to find out how knowledgeable a student is in the topics being evaluated. The curriculum graph in Fig. 1. has been altered in Fig. 2. to graphically illustrate each vertex's out-degree metric. In Fig. 2., the out-degree metric for each node in the graph was determined using Gephi Software. The size and color of the vertices were then automatically scaled based on these values, making nodes with low out-degrees appear little with light green color and nodes with high out-degrees appear bigger with dark green color.

For example, vertex ELEC320 has an out-degree measure of 0; consequently, it is one of the smallest nodes in the graph. Vertex MATH130 has an out-degree of 4 which is the largest out-degree in the curriculum and is displayed as the largest vertex on the graph with dark green color. The other vertices have different size based on the metric value. Fig. 2. shows that MATH130 and PHYS105 are the optimal places to conduct baseline assessment and introduce topics. While, ELEC 320 will be the worst place.

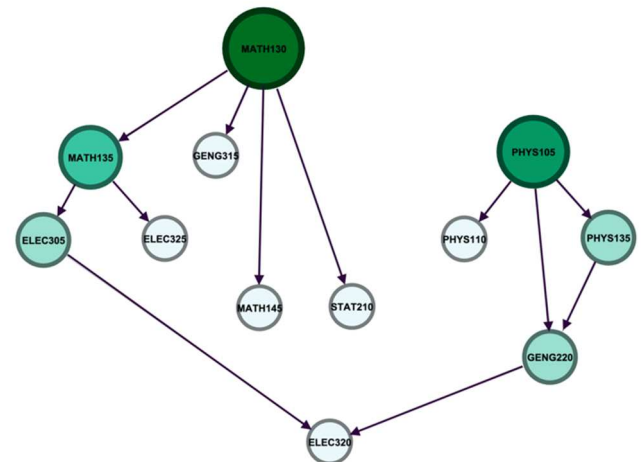


Fig. 2. Sample of the electrical engineering curriculum at UAEU - Out-Degree emphasized.

2) *In-Degree*: The vertex in-degree is the number of directed edges coming into the vertex. It serves as both; a clue as to the network flow's final destination, and a crucial point of exposure for that result of the network flow. Fig. 3. displays the in-degree for each vertex via the node size and color.

The in-degree measure, used in the context of the curriculum, identifies courses where students should participate in activities for higher-level learning, such as tasks that require evaluation, synthesis, or analysis. Additionally, it highlights the best courses for final assessment at a specific program.

Fig. 3. shows that ELEC320 and GENG220 are good places for assessment. GENG220 should be selected for intermediate topic assessment while ELEC320 should be selected for full program assessment.

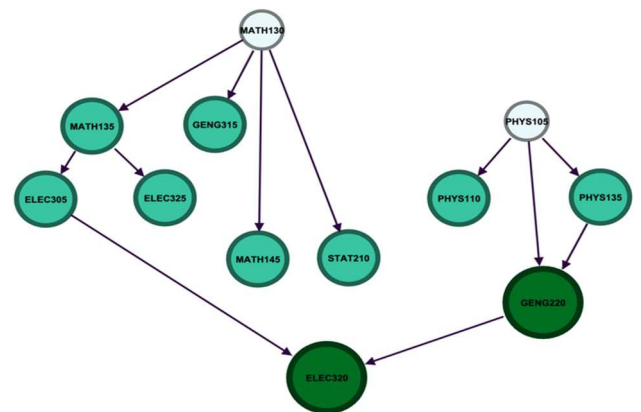


Fig. 3. Sample of the electrical engineering curriculum at UAEU - In-Degree emphasized.

3) *Betweenness Centrality*: Finding vertices that connect one area of a graph to another can be done by using the betweenness centrality measure. The course with the highest betweenness centrality value within a curriculum setting denotes a crucial connection between program tracks or course clusters. ELEC320 course is expected to have the highest betweenness centrality within the curriculum in Fig. 3. It was noticed that betweenness centrality and in-degree measure indicate that ELEC320 should be selected for full program assessment.

B. Curriculum Design and Restructure

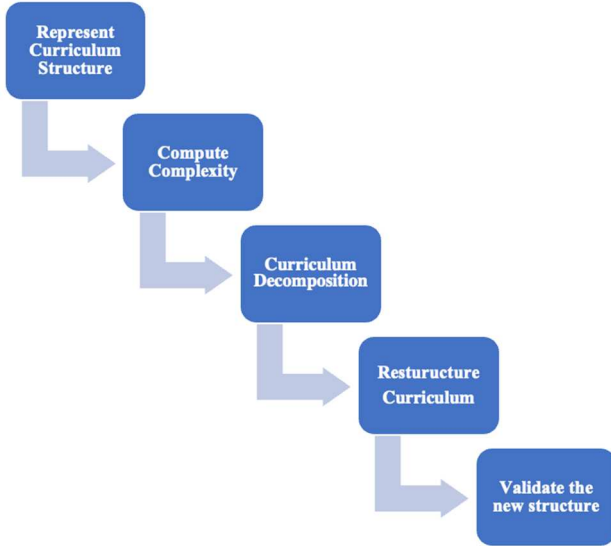


Fig. 4. Steps of curriculum restructuring.

Fig. 4. displays the steps used to adjust the curricular structure and design to enhance the graduation rate. It starts by presenting the courses of the curriculum and the relationship between them as a graph consisting of nodes and edges. Next, both delay factor and blocking factor for each course will be generated and the summation of them generates the complexity of each course. The course with the highest complexity indicates a critical course that is better to be restructured in the curriculum design to reduce its complexity and enhance the graduation rate. Then, the learning outcomes and prerequisites of the complex curricular patterns will be analyzed. After that, the common learning outcomes and the prerequisites of the complex curricular patterns will be used to generate a new structure for the courses that will have low complexity leading to a higher graduation rate. Finally, the new structure will be validated by computing its complexity. The new structure is expected to have lower complexity which will enhance the students' graduation rate.

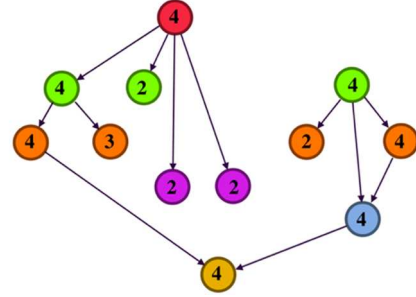
In this section, the calculation of the delay factor, blocking factor, and complexity for the curriculum sample in Fig. 1. is illustrated.

A curriculum's graph structure contains two helpful metrics. First, the blocking factor of a course which is the number of subsequent courses that a student cannot enroll in before passing a that course. Second, the delay factor which is the total number of courses on any prerequisite pathway that contains that course.

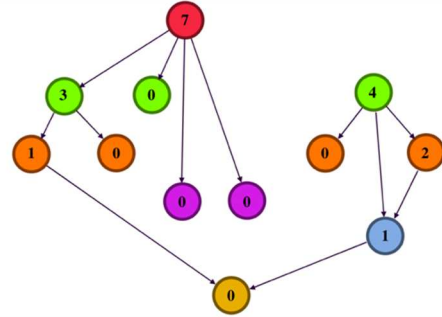
The curriculum displayed in Fig. 5. is made up of twelve courses. Fig. 5. (a) shows the delay factor for each node while

Fig. 5. (b) shows the blocking factor for each node. Fig. 5. (c) displays the complexity of each node which is the summation of the delay factor and the blocking factor. The structural complexity of a pattern is defined as the sum of the courses' complexity in a pattern [18]. It is noticed that the structural complexity of a curriculum affects students' ability to successfully progress through the curriculum. Identifying the courses with the highest cruciality or complexity is a critical step to enhance the structural design and improve the students' graduation rate [19].

(a)



(b)



(c)

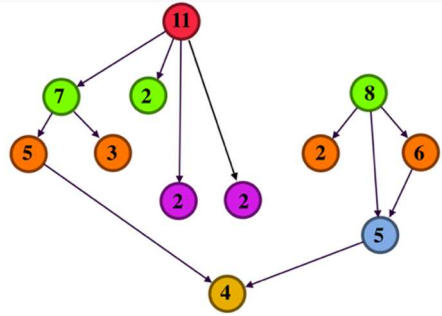


Fig. 5. (a) the delay factor (longest path) for each course, (b) the blocking factor (connectivity) for each course, and (c) the complexity or cruciality of each course (complexity = connectivity + longest path).

IV. CONCLUSION

The method created and discussed in this work transformed the curriculum issue into a graph, an abstract mathematical representation. The curriculum structure was then examined using graph-theoretic metrics in order to determine the best courses for introducing new topics, reinforcing existing topics, and introducing changes to topics. The measurements also showed where in the curriculum primary and secondary assessments should be placed. The method is simple to use and inexpensive because it makes use of Gephi free software. The method also includes a

configurable component for visualization that creates a picture of the curriculum that is simple for non-mathematicians to comprehend and explain. The approach was implemented to a real portion of United Arab Emirate University electrical engineering curriculum and proved to provide outcomes that are consistent. Moreover, course cruciality was generated via calculating both delay factor and blocking factor. These measurements were used to consider possible enhancement in graduation rates by conducting curriculum restructuring.

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